- Chapter 2 introduces the fundamental concepts of reinforcement learning that will reappear through the rest of the book. We also implement our first practical reinforcement learning algorithm.
- Chapter 3 introduces deep Q-learning, one of the two broad classes of deep reinforcement algorithms. This is the algorithm that DeepMind used to outperform humans at many Atari 2600 games in 2015.
- Chapter 4 describes the other major class of deep reinforcement learning algorithms, policy-gradient methods. We use this to train an algorithm to play a simple game.
- Chapter 5 shows how we can combine deep Q-learning from chapter 3 and policy-gradient methods from chapter 4 into a combined class of algorithms called actor-critic algorithms.

Part 2 builds on the foundations we built in part 1 to cover the biggest advances in deep reinforcement learning in recent years.

- Chapter 6 shows how to implement evolutionary algorithms, which use principles of biological evolution, to train neural networks.
- Chapter 7 describes a method to significantly improve the performance of deep Q-learning by incorporating probabilistic concepts.
- Chapter 8 introduces a way to give reinforcement learning algorithms a sense of curiosity to explore their environments without any external cues.
- Chapter 9 shows how to extend what we have learned in training single agent reinforcement learning algorithms into systems that have multiple interacting agents.
- Chapter 10 describes how to make deep reinforcement learning algorithms more interpretable and efficient by using attention mechanisms.
- Chapter 11 concludes the book by discussing all the exciting areas in deep reinforcement learning we didn't have the space to cover but that you may be interested in.

The chapters in part 1 should be read in order, as each chapter builds on the concepts in the previous chapter. The chapters in part 2 can more or less be approached in any order, although we still recommend reading them in order.